

Gaussian Type Orbitals basis sets for the calculation of continuum properties in molecules: the differential photoionization cross section of acetylene

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Abstract

The integral and the differential cross sections of the valence one-photon ionization of C_2H_2 are computed in the Random Phase Approximation using large L^2 basis sets of Gaussian Type Orbitals (GTO's) and a K-matrix based technique. The use of special Polynomial Spherical GTO's allows an adequate representation of the short range part of the orbitals lying in the electronic continuum. Ionization channels originating from the $1\pi_u$, $3\sigma_g$, $2\sigma_u$ and $2\sigma_g$ occupied orbitals are considered, both in Separated and Interacting Channel approaches. An accurate account of the effect of the interaction between the different ionization channels is found to be mandatory in order to reproduce some of the features observed in photoelectron experimental spectra. © 2000 Elsevier Science B.V. All rights reserved.

1. Introduction

In the last few decades acetylene has been the subject of an extensive experimental work in photoelectron spectroscopy laboratories. The relative simplicity of the system, coupled with its physical characteristics, which make handling and storing quite safe, are such that acetylene, together with its iso-electronic analogues N_2 and CO, is an ideal candidate for accurate benchmark spectroscopic studies.

Also, its dynamic behavior upon excitation beyond the first ionization potential, the existence of shape resonance structures and the competition between neutral dissociation and autoionization are all issues that make the system one of the most attractive for experimentalists.

A detailed account of the work done since the first pioneering work of Baker and Turner [1] and a long list of references is given for instance by Cooper et al. in Ref. [2], by Avaldi et al. [3] and, very recently, by Holland and coworkers in Ref. [4].

The absolute photoabsorption cross sections of C_2H_2 have been measured in Refs. [5–10]. Several Rydberg series with associated vibrational progressions were detected in studies at photon energies less

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than 11.8 eV [6,7,11–14]. Data in the windowless VUV region from 12.4 eV to 70.85 eV are found in Refs. [5,9,15]. Photoionization cross sections were measured in Refs. [16–23]. Photoelectron spectra have been obtained using either selected atomic resonance lines [23–32] or synchrotron radiation [3,18,21,22,33–41]. Low impact energy electron energy loss studies were performed in ranges from 5 eV to 15 eV [42,43]. The carbon 1s energy loss spectrum of C_2H_2 was also studied in detail [44–46]. Relative yields (cross sections) curves as function of photon energy have been measured for the different parent ions, see Refs. [2,20].

The authors of Ref. [2] use dipole E-loss spectroscopy to obtain absolute photoabsorption cross sections of C_2H_2 from 6 to 80 eV, and the (e,e + ion) technique to measure photofragmentation ion branching ratios and partial oscillator strengths (or f-values) for molecular and all dissociative photoionization channels from the ionization potential up to 80 eV equivalent photon energy. Absolute data were obtained by normalizing the absorption area from 0 to infinity to 10.3 electrons.

Avaldi and co-workers [3] present an high resolution photoelectron spectrum of acetylene obtained in the 11.3 to 20 eV energy region by threshold photoelectron spectroscopy and constant photoelectron energy spectroscopy. Holland and co-workers [4] have studied the dynamics of the valence shell photoionization of C_2H_2 , in the photon energy range 13–140 eV, using photoelectron spectroscopy and synchrotron radiation.

In spite of the noticeable amount of work on acetylene, the study of the photoelectron angular distribution has received surprisingly much less attention. Prior to Ref. [4], the only data available could be found in Refs. [21,25,26,35,36].

Equally few are the examples of theoretical studies of the photoionization partial cross sections and of the photoelectron asymmetry parameters of acetylene. Rydberg excitation energies and photoabsorption cross sections of C_2H_2 were studied theoretically, using a Static-Exchange approach, up to 30 eV [47]. A similar Static-Exchange study is that by Langhoff and co-workers in Refs. [18], where, as in Ref. [47], the Stieltjes-Tchebycheff moment theory is employed to obtain a continuum photoionization profile from a discrete L^2 oscillator strength distribu-

tion. In the work of Lynch and co-workers [48] the cross sections were obtained from frozen-core Hartree-Fock continuum orbitals, and the Iterative Schwinger variational method was employed to solve the self-consistent equations in the continuum. A continuum multiple scattering model was employed in Refs. [26] and [35]. The partial cross section of all the σ_g orbitals of acetylene was studied by Farren et al. [49] with the aim to correlate the prominent features of the spectrum with the presence of Mulliken virtual valence orbitals. The inner valence photoionization spectrum of acetylene was also the subject of some recent work of Lucchese and co-workers [50,51].

The electronic configuration of C_2H_2 is

$$(1\sigma_g)^2(1\sigma_u)^2(2\sigma_g)^2(2\sigma_u)^2(3\sigma_g)^2(1\pi_u)^4 \\ \times (1\pi_g)^0(3\sigma_u)^0 \quad (1)$$

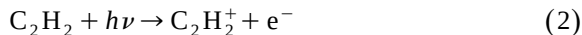
for a $^1\Sigma_g^+$ state. The adiabatic ionization energies were measured to be of 11.403 eV (for the $C_2H_2^+ X^2\Pi_u$), 16.297 eV ($A^2\Sigma_g^+$) and 18.391 eV ($B^2\Sigma_u^+$), as taken from Ref. [23]. High-resolution photoelectron spectroscopy (PES) studies [29] place the thresholds at 11.40 eV (to the ionic state $X^2\Pi_u$), 16.38 eV (to the $A^2\Sigma_g^+$), 18.38 eV (to the $B^2\Sigma_u^+$) and 23.5 eV (to the $C^2\Sigma_g^+$). The K edge occurs at 291.1 eV [45,52].

In the last fifteen years our group has developed a good expertise in the treatment of the properties of the continuum in molecules with L^2 techniques [53–55]. The details of the differential cross section are obtained through a method based upon the determination of the reaction K-matrix in a version adapted to the use of L^2 basis sets [56]. This approach can be implemented at different degrees of sophistication of the wave function: SCF based approximation [56], leading to the Static Exchange Approximation, accurate CI type [57,58] and also the RPA to obtain directly the relevant transition matrix elements [59]. The latter is employed here.

In the last few years we have devoted our attention to the efficient use of Gaussian Type Orbitals (GTO's) as expansion sets for the molecular continuum orbitals. Applications have been made to the photoionization of H_2 [60], Li_2 and LiH [61] and N_2 [62]. By using a special K-matrix approach and suitable Polynomial Spherical GTO's (PSGTO) [63],

the limitations of usual GTO's and their inadequacy to describe Coulomb waves at large distances from the origin were overcome. The advantages of the use of PSGTO's in molecular photoionization are numerous; among others, we mention here the possibility of exploiting computer codes written for the study of bound states, as the multicenter one- and two-electron integrals program developed in our group [64].

In this paper the process of interest is



with the range of energies extending in the continuum up to 100 eV, well beyond the inner valence ionization threshold e.g. for the photoejection of electrons from the four valence molecular orbitals ($1\pi_u$, $3\sigma_g$, $2\sigma_u$ and $2\sigma_g$).

In the following, after a brief review of the theory and a description of the computational tools, a detailed account of the features of the differential photoionization cross section profile of C_2H_2 will be given.

2. Theory

The theoretical method and the computational algorithm employed have been fully described elsewhere [53–55,62]; thus here only a general outline is given. Our method to obtain the states into the continuum may be considered a Configuration Interaction approach where the multielectron basis, formally, includes the one-electron continuum. The diagonalization of the resulting energy matrix [65] is equivalent to the solution of the system of integral equations for the half-on-the-shell reaction K-matrix. For each symmetry Γ the asymptotic continua are represented as linear combinations of partial wave channel (*pwc*) basis functions $|\Phi_{\gamma,E}\rangle$ built as products of well defined ion states $|\Phi_{E_I,S_I,\Gamma_I}^{N-1}\rangle$ times continuous orbitals $\varphi_{l,m,\epsilon}$, representing in practice very diffuse functions, extending far beyond the molecular region. These products are coupled to give the correct total symmetry Γ and spin multiplicity S . A Greek index collects all relevant labels i.e. $\gamma \equiv E_{I,\gamma}, S_{I,\gamma}, l_{\gamma}, m_{\gamma}, S_{\gamma}$ with $E = E_{I,\gamma} + \epsilon$.

In addition to the *pwc*'s the basis includes also a “localized channel” (*lc*) set of functions $|\Phi_{C,E_j}\rangle$, linear combination of Slater determinants built from a set of localized (short range) orbitals. The multi-

electron basis diagonalizes the electronic molecular hamiltonian separately for each channel and for each ion state considered, i.e.:

$$\langle \Phi_{C,E_j} | \hat{H} | \Phi_{C,E_k} \rangle = \delta_{j,k} E_j, \quad \langle \Phi_{C,E_j} | \Phi_{C,E_k} \rangle = \delta_{j,k}, \quad (3)$$

$$\langle \Phi_{\beta,E''} | \hat{H} | \Phi_{\beta,E'} \rangle = \delta_{E'',E'} E'', \quad \langle \Phi_{\beta,E''} | \Phi_{\beta,E'} \rangle = \delta_{E'',E'}, \quad (4)$$

$$\langle \Phi_I^{N-1} | \hat{H}^{N-1} | \Phi_I^{N-1} \rangle = \delta_{I',I} E_I, \quad \langle \Phi_I^{N-1} | \Phi_I^{N-1} \rangle = \delta_{I',I}, \quad (5)$$

where the $\delta_{E'',E'}$ stands both for a Dirac $\delta(E'' - E')$ and a Kronecker symbol, depending on the spectral region. Thus while the resulting *pwc*'s give a satisfactory representation of the asymptotic continua, the *lc* accounts for the localized discrete and autoionizing states and it supplies a suitable basis for the description of correlation between the photoelectron and its parent ion. It should be noted that bases of different channels belonging to the same symmetry will not be orthogonal. Furthermore the diagonalization of the *pwc* may produce some discrete states which lie in the continuum of other channels; these will be labeled as pseudo-discrete states.

Employing the full basis, comprehensive of all *pwc*'s and the *lc*, the stationary states in the continuum $|\Psi_{\beta,E}\rangle$ may be expressed as

$$|\Psi_{\beta,E}\rangle = |\Phi_{\beta,E}\rangle + \sum_{\gamma} \mathcal{Z} dE' |\Phi_{\gamma,E'}\rangle \frac{\mathcal{P}}{E - E'} K_{\gamma E', \beta E} \quad (6)$$

where the symbol $\mathcal{Z}$ stands for summation upon the discrete states and integration upon the energy in the continuum. The sum upon the channels includes, besides all *pwc*'s, also the *lc*. The requirement

$$\langle \Phi_{\theta,E''} | \hat{H} - E | \Psi_{\beta,E} \rangle = 0; \quad \forall \theta, E'' \quad (7)$$

by defining the matrix elements $V_{\theta E'', \gamma E'}(E)$ as

$$\langle \Phi_{\theta,E''} | \hat{H} - E | \Phi_{\gamma,E'} \rangle = \delta_{\theta,\gamma} \delta_{E'',E'} (E'' - E) + V_{\theta E'', \gamma E'}(E) \quad (8)$$

yields the system of integral equations

$$\sum_{\gamma} \mathcal{Z} dE' \left\{ \delta_{\theta,\gamma} \delta_{E'',E'} - V_{\theta E'', \gamma E'} \frac{\mathcal{P}}{E - E'} \right\} K_{\gamma E', \beta E} = V_{\theta E'', \beta E} \quad (9)$$

for the $K_{\gamma E', \beta E}$ coefficients, which are the elements of the half-on-the-shell K-matrix. The states $|\Psi_{\beta, E}^{(-)}\rangle$ satisfying the boundary conditions appropriate for a photoionization process are given by

$$|\Psi_{\beta, E}^{(-)}\rangle = \sum_{\gamma} |\Psi_{\gamma, E}\rangle [\mathbf{1} - i\pi \mathbf{K}(E)]_{\gamma, \beta}^{-1} \quad (10)$$

where $\mathbf{K}(E)$ is the matrix upon the energy shell. It should be noted that Eq. (9) for the half-on-the-shell K-matrix insure that it is stationary on-the-shell [62]. Since the K-matrix is related to the S-matrix by

$$\mathbf{S}(E) = \frac{\mathbf{1} - i\pi \mathbf{K}(E)}{\mathbf{1} + i\pi \mathbf{K}(E)}$$

the positions and the widths of the autoionizing states may then be found by looking for the zeros of $\mathbf{1} + i\pi \mathbf{K}(E)$ via analytical continuation.

The use of L^2 orbitals implies that the diagonalization of the electronic molecular hamiltonian projected upon our multielectron basis cannot yield the true continuum; only a discretized representation of the system of integral Eqs. (9) is available. The grid points correspond to the energy eigenvalues yielded by the separate channel diagonalization. An important point for the success of the method is the knowledge of the additional phase shifts $\delta(k)$ of the orbitals $\varphi_{l, m, \epsilon}$ with respect to the true Coulomb waves. These are needed to evaluate the asymmetry parameters and can be determined by employing the special technique described elsewhere [61], a technique which is made especially successful by the inclusion of suitable PSGTO's in the basis used to represent the continuum orbitals.

It is worthwhile to emphasize that the $\langle \Phi_{\theta, E''} | (E - \hat{H}) | \Phi_{\gamma, E'} \rangle$ matrix elements are evaluated without making any approximation for the lack of orthogonality among different channels. Their exact evaluation, together with the presence of the l_c , are of extreme importance for the success of the technique. These ingredients account properly for the correct behavior of the continuum states in the molecular region. The use of high values of l , which appears to be mandatory in other methods [66,67], is thus unnecessary in our case. Moreover, the feature is extremely convenient from a computational viewpoint. The approach outlined above can be implemented at different degree of sophistication of the wave function, starting from the SCF based approximation [57]

to the more accurate CI type [57]. In addition it may also be applied to the EOM method [62] in order to obtain directly the transition matrix elements $\rho_{0, \beta E}$ involving the electronic continua. In order to satisfy the boundary conditions for photoionization, analogously to Eq. (10), the real $\rho_{0, \beta E}$ should be transformed according to

$$\rho_{0, \alpha E}^{(-)} = \sum_{\beta} \rho_{0, \beta E} [\mathbf{1} - i\pi \mathbf{K}(E)]_{\beta, \alpha}^{-1}, \quad (11)$$

where as usual $\mathbf{K}(E)$ is the on-the-shell RPA K-matrix [62].

In this paper the RPA has been employed either in the Single Channel (SC) approximation – where only one of the occupied orbitals is emptied – and in the Interacting Channel (IC) approach – where all considered channels are allowed to interact. We label our approach “KM-RPA”.

3. Computational details

The KM-RPA method has been applied to the calculation of the partial cross section and of the asymmetry parameter β

$$\frac{d\sigma_{j_\gamma}(\omega)}{d\hat{k}} = \frac{\sigma_{j_\gamma}(\omega)}{4\pi} [1 + \beta_{j_\gamma} P_2(\cos\theta)] \quad (12)$$

of the outer valence $j_\gamma \equiv (1\pi_u)^{-1}, (3\sigma_g)^{-1}$ and $(2\sigma_u)^{-1}$ and the inner valence $(2\sigma_g)^{-1}$ photoionization channels of C_2H_2 for linearly polarized photons with energy up to 100 eV. The general computational scheme was already outlined in Ref. [61]; we give here only a few comments concerning the peculiarities of the molecule under study.

The experimental equilibrium geometry (C–C = 1.2 Å; C–H = 1.06 Å) was assumed. The molecule was placed along the z -axis with the center of nuclear mass in the origin. The SCF ground state of the neutral molecule was described employing the 6-31G* basis set [68] with the valence GTO's functions fully decontracted and the exponent of the d function variationally determined ($\xi = 0.84$). The resulting energy was $E_{SCF} = -76.820255$ Hartree; Koopman's ionization potentials are reported in Table 1, and compared with experiment as taken from Ref. [2]. To make the comparison easier, the latter were

Table 1
Koopman's and experimental [2] valence ionization potentials (eV) of C_2H_2

Assignment	Koopman	Experiment
X $1\pi_u$	10.971	11.40
A $3\sigma_g$	18.344	16.38
B $2\sigma_u$	20.739	18.38
C $2\sigma_g$	27.843	23.60

employed in the KM-RPA calculations. Excitations involving all four valence orbitals were all included in the KM-RPA calculations.

The *pwc* orbitals for a given occupied SCF orbital and for the corresponding (l, m)'s were determined by solving the one-electron problem yielded by the appropriate SEA Hamiltonian. They were expanded on GTO's and on PSGTO's centered at the midpoint of the molecular axis. These functions were proven to be particularly efficient to represent orbitals lying in the continuum for molecular systems [61–63]. In particular, in Ref. [63] it was shown that, using $X\alpha$ local potentials, SC cross sections of high accuracy (within 2–3%) can be obtained with a suitable choice of the PSGTO's.

The maximum value of l here is 7. A detailed analysis of the convergence characteristics of both the integral and the differential cross section with the angular quantum number has been given in Ref. [62] for the molecule of N_2 . These convergence requirements may be tougher for β than for the integral cross section. In the present study we made some exploratory calculations employing *pwc* sets with l up to 5. The changes in β with respect to the case of l up to 7 were small, such as to lead us to conclude that the approximation introduced by truncating the partial wave expansion to $l = 7$ is far less questionable than any of the other possible causes of inaccuracy in the calculation.

In order to obtain wave functions of the correct spin-spatial symmetry, linear combinations of singly-excited detors were taken, as explained in detail in our previous investigation on N_2 [62]. The calculations involved finally a total of 15 *pwc*'s for the $^1\Sigma_u^+$ final state symmetry, and of 18 *pwc*'s for each component of the $^1\Pi_u$ final state symmetry.

The *lc*'s include excitations to the virtual SCF orbitals and to a few tight functions with l up to 3.

The latter are included for a better description of the behavior of the final one-electron continuum states near the nuclei, where the dip of the potential localizes some electronic density. It should also be emphasized that the *lc*'s consist essentially of excitations to discrete or pseudo-discrete orbitals, and of very broad wavepackets lying in the continuum of the considered channels. Moreover they allow to take into account some effects arising from the incompleteness of the partial wave expansion [61]. Differently from the *pwc* orbitals, the *lc* orbitals are orthogonalized to the occupied SCF orbitals.

To deal adequately with the numerical problems arising from the redundancy of the complete virtual orbital space, appropriate basis set transformations are performed before the K-matrix calculation [69]. As described in some details in Ref. [62], such transformations can produce unphysical Feshbach-type resonances whose effects on the resulting spectrum needs to be considered carefully. It is seen that, as for N_2 , the “intruder” states exhibit very large linewidths, their effect being spread smoothly over a wide energy range. The transformations do not introduce apparently spurious artifacts in the spectrum resulting from the K-matrix calculations.

The total and partial cross sections were convoluted with a Gaussian function with Full Width of Half Maximum (FWHM) of 1.5 eV, a value chosen as a compromise between the Franck-Condon amplitudes of the valence ion states and the spectral resolution of the available experimental data. The asymmetry parameters β were convoluted with the same Gaussian function times the corresponding partial cross section.

We will report both the Interacting Channel (IC-RPA) and Single Channel (SC-RPA) results, aiming to extract useful information on the photoionization dynamics.

4. Results and discussion

4.1. Total cross section

The integral photoabsorption cross section of C_2H_2 computed in the KM-RPA is shown in Fig. 1. The four curves correspond to our results in both the IC and SC approximations and in the Length (L) and Velocity (V) forms of the interaction operator. The

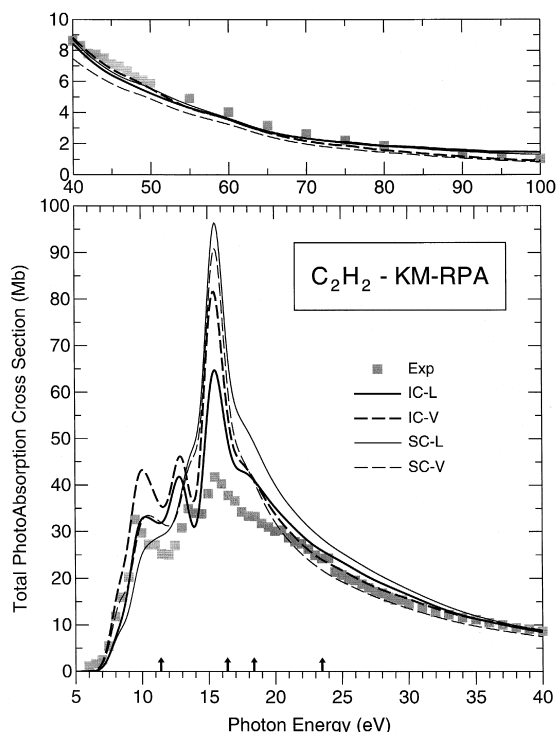


Fig. 1. The total photoabsorption cross section of C_2H_2 (Mb) in the 5 eV to 40 eV photon energy range (below) and 40 eV to 100 eV (above). KM-RPA results, in the Interacting Channel (IC) and Separated Channels (SC) approximation both for the Length (L) and Velocity (V) Gauge. Also reported are experimental data taken from Ref. [2]. Arrows on the photon energy axis indicate the threshold for each of the four channels X, A, B, C. Here and in the following figures our K-matrix results are convoluted with a normalized Gaussian function with FWHM of 1.5 eV.

IC total cross section has been obtained as the sum of the four valence partial cross sections, by including the f -values of the discrete excited states lying below the first ionization threshold. Analogously, in the SC case the f -values of the pseudo-discrete states lying in the discrete region of each non-interacting channel – and which in the IC approximation may mix with the continuum of other channels – were included as well. Both for IC and SC, these f -values are convoluted with a Gaussian function of width 1.5 eV before being added to the oscillator strength arising from the continuum.

In Fig. 2 the total photoionization cross section up to 30 eV is displayed. It is evaluated by neglecting all the intensities for transitions to discrete states in

the IC approximation and all the intensities for transitions to discrete and pseudo-discrete states in the SC. In Fig. 3 the unconvoluted total ionization cross section is reported for both dipole allowed symmetries of the final ionized state. The Rydberg series embedded in the continuum and converging to the A and B ionic states are particularly evident.

At photon energies higher than 24 eV (Fig. 1), where all valence channels are open, both the IC and the SC cross sections are smooth and closely match the E-loss data [2]. In this region the inclusion of the channel interaction improves considerably the gauge agreement without altering the general behaviour of the cross section. Therefore RPA seems to be adequate for the description of the states lying in this energy region.

For photon energies lower than 24 eV, both the theoretical and the experimental cross sections show three pronounced peaks.

(i) The first peak is found in the discrete region (≈ 10 eV) and it is mainly originated from $1\pi_u \rightarrow 4\sigma_g$ (8.3 eV) and $1\pi_u \rightarrow 1\pi_g$ (9.6 eV) excitations. Other less intense transitions in the discrete are

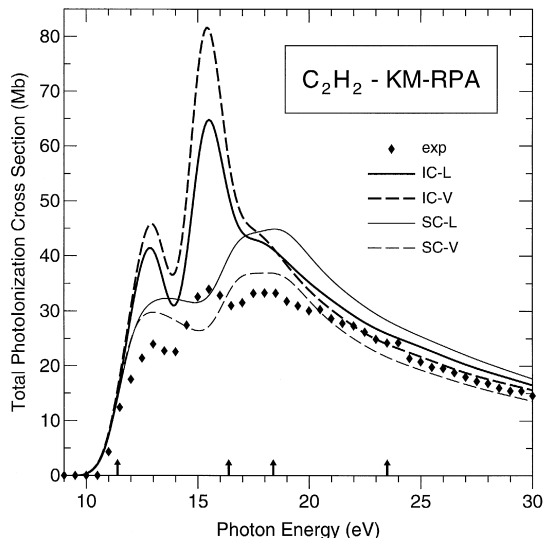


Fig. 2. The total photoionization cross section of C_2H_2 (Mb) in the 10 eV to 30 eV photon energy range. The cross section is convoluted with a Gaussian function (see text). Also reported are the experimental data taken from Ref. [2]. Arrows on the photon energy axis indicate the threshold for each of the four partial channels.

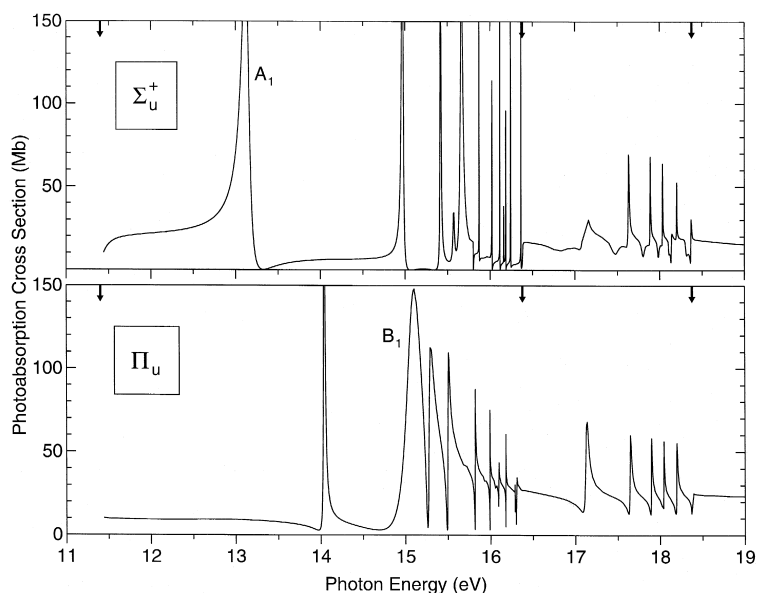


Fig. 3. The total (unconvoluted) photoionization cross section of C_2H_2 (Mb) up to the third ionization threshold for the two dipole allowed symmetries. IC-RPA, Length gauge. Arrows on the photon energy axis indicate the ionization thresholds of the X, A and B partial channels. The A_1 and B_1 labels in the figure refers to the two intense Feshbach resonances discussed in the text.

found above 10 eV. Using a smaller FWHM for the convolution, three peaks at 8.3, 9.6 and 10.5 eV would appear.

(ii) The second peak corresponds to the Feshbach resonance A_1 $3\sigma_g \rightarrow 3\sigma_u(^1\Sigma_u^+)$ at about 13 eV (see also Fig. 3). The excited orbital $3\sigma_u$ has been identified as Rydberg orbital of p character [18] possibly with some antibonding components [41]. In the SC case this transition energy is found at slightly higher energy (13.7 eV) and it appears as a shoulder in the SC convoluted cross section. The corresponding molecular transition $3\sigma_g \rightarrow \sigma_u^*(^1\Sigma_u^+)$ is found near the $3\sigma_g$ threshold.

(iii) The third and most intense peak arises from a second Feshbach resonance B_1 $2\sigma_u \rightarrow \pi_g^*(^1\Pi_u)$. In the unconvoluted Π_u cross section of Fig. 3 this peak is visible at ~ 15.1 eV and mixes with the less intense Rydberg series converging to the A threshold. Comparison of the SC and IC results shows that this mixing perturbs the Rydberg series and strongly increases its intensity. It is worthwhile to notice the different lineshape of the B_1 and of the Rydberg bands in Fig. 3.

The origin of the second and third peak is confirmed by Fig. 2 where the A_1 and B_1 intensities do

not contribute to the SC cross section. Therefore the difference between the IC and SC cross sections is due here mainly to the contribution of these resonances. The SC photoionization cross section is everywhere smooth and much closer to the experimental data with respect to the IC cross section.

In this energy region the agreement between both the SC and IC absorption cross sections and the experiment is rather poor. Although the position of the second and third peak is close to the E-loss maxima (13.3 and 15.5 eV for A_1 and B_1 respectively [4]), the IC intensity is too high even in comparison with the optical data of Ref. [5] which show a maximum of 50 Mb for the B_1 band. Our f -values for both the A_1 and B_1 resonances in the SC case are very close to those obtained in the Static-Exchange approximation by Langhoff and co-workers [18,47].

For the B_1 resonance, where major discrepancies are found, we observe that going from SC to IC there is a lowering of the intensity without any improvement in the gauge agreement. Incidentally we remark that in the SC calculation the L gauge gives the largest intensity, while the reverse is true for the IC case.

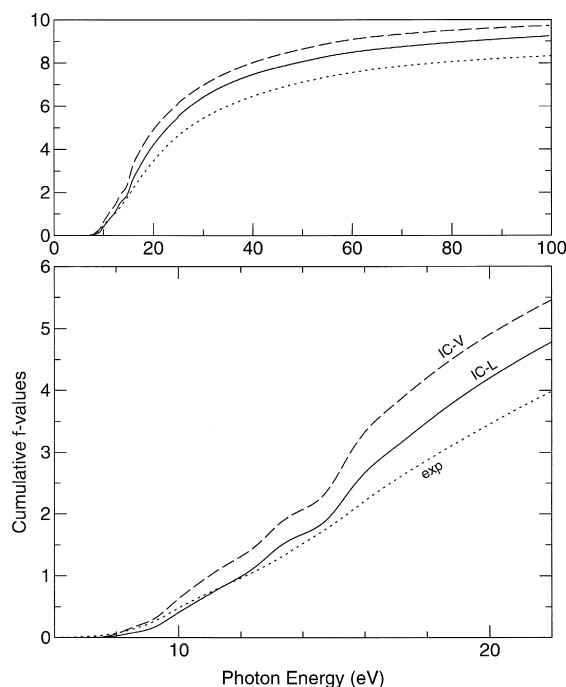


Fig. 4. The cumulative oscillator strength of C_2H_2 up to 22 eV photon energy range (below) and up to 100 eV (above). Experimental results [2] are compared to Interacting Channel RPA results for both the Length (L) and Velocity (V) choices of Gauge for the interacting operator.

Further information can be obtained by considering the cumulative f -value function for both theoretical and experimental (Ref. [2]) data, as reported in Fig. 4 for the absorption cross section. In coincidence with the B_1 resonance the disagreement between the theoretical and experimental curves increases quite rapidly by about 0.3. Consistently with the close agreement between theory and experiment for higher photon energies, this difference does not change appreciably at least up to 100 eV. The total area of the E-loss cross section is of 10.3, as estimated by the authors of Ref. [2]. That of the theoretical curves is 9.9 and 10.8 for the L and V gauges, respectively. Thus from the value of the curves at the right edge in Fig. 4, it appears that the residual area from 100 eV onward is of 0.7, 1.1 and ≈ 2 for L, V and E-loss, respectively. Thus beyond 100 eV our cross sections on average should be lower than those observed in the E-loss experiment.

These considerations lead to the conclusion that our results would better match experiment if, by

using a more refined approximation, some oscillator strength could be displaced to energies higher than 100 eV. It is our opinion that both the local gauge disagreement and the energy delocalization of the f -values of the A_1 and B_1 autoionizing states, would be improved if suitable double excited configurations were included in the final state space.

4.2. Photoionization from the $1\pi_u$ orbital

Fig. 5 displays our results for the partial photoionization cross section of acetylene for a process leading to the ionic $X^2\Pi_u$ state. Separated channel and Interacting channel results are compared to the (e,e + ion) experimental data of [2].

The interpretation of the spectrum of acetylene below 20 eV is complex and it is influenced by a

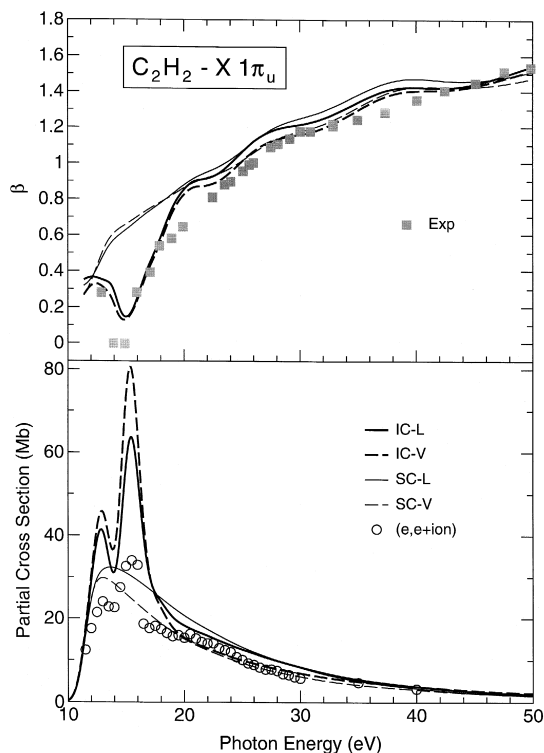


Fig. 5. The partial $(1\pi_u)^{-1}$ photoionization cross section of C_2H_2 (Mb) in the 10 eV to 50 eV photon energy range (below). The asymmetry parameter β is shown in the upper section of the figure. Also reported are the experimental (e,e + ion) data extracted from Ref. [2] and the PES photoelectron angular distribution results of the measurements of Holland and co-workers [4].

number of autoionizing resonances. There is in some cases disagreement on the assignments of some of its features, and if it is now generally accepted that the $B_1 (2\sigma_u \rightarrow \pi_g^*)$ transition is responsible for the resonantly enhanced main peak at ≈ 15 eV, different authors have proposed different mechanisms to explain the other fine details of the spectrum, see for instance Refs. [4,41,70].

As already seen in Section 4.1, the effect of the interaction between the ionization channels has a striking evidence in the noticeable difference between the cross section computed in the SC- and IC-approximations in the energy region below 20 eV. Two distinct well defined peaks appear as soon as we switch the coupling on: a lower peak around 13.5 eV, due to the bonding-antibonding Feshbach resonance A_1 and the stronger peak at about 15 eV due to the B_1 resonance already discussed for the total cross section [21]. The structure is clearly visible in the experimental spectrum (see also Refs. [18] and [35]), but the calculated intensity is almost twice that observed. We have discussed above the origin of this enhancement in our low energy spectrum. The peaks are clearly absent in the SC approximation. Levine and Soven [71] have done time-dependent local-density-approximation calculations which include channel interactions and they could account for the features of partial cross section in this region.

The asymmetry parameter β for the X $(1\pi_u)^{-1}$ channel is shown on the upper portion of Fig. 5, and in great detail in a reduced energy region in Fig. 6. In Fig. 5 the SC- and IC-approximation results are compared with the very recent data recorded by Holland and co-workers [4]. The remarkable effect of the channel interaction becomes if possible even more evident. No sign of the deep observed by experimentalists and appearing around 15 eV, e.g. where the cross section shows a pronounced maximum, can be seen in the SC- results. The vibrationally unresolved experimental deep goes as low as 0, corresponding to a completely isotropic angular distribution. After convolution, our calculations estimate a minimum of $\beta \approx 0.1$, and a width of ≈ 1 eV, see main body of Fig. 6. The minimum, related clearly to Feshbach resonances – B_1 and the progression converging to the ionic state A – is reproduced in the upper part of the Figure, where the unconvoluted asymmetry parameter is shown. The deep was

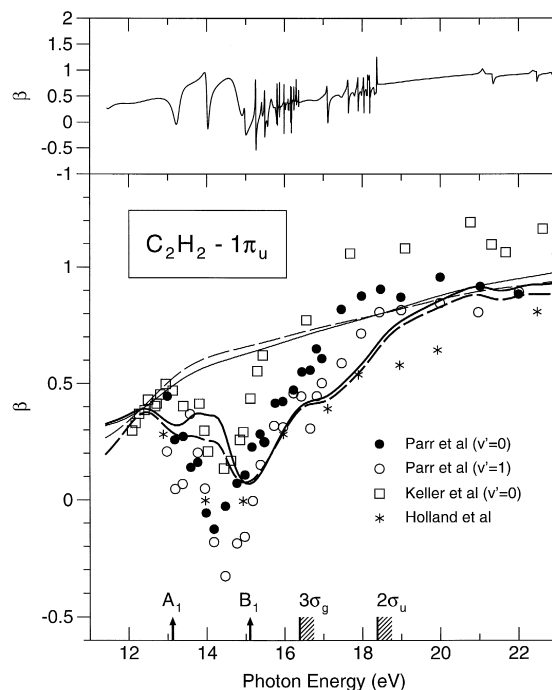


Fig. 6. The details of the asymmetry parameter β for the partial wave channel $(1\pi_u)^{-1}$ in C_2H_2 in the 11 eV to 23 eV photon energy range. In the main body of the figure the KM-RPA results, in the Interacting Channel (IC) (heavy lines) and Separated Channel (SC) (light lines) approximations both for the Length (solid line) and Velocity (dash line) choice of Gauge for the interacting operator are shown. The Feshbach resonances A_1 and B_1 (arrows) and the A and B thresholds are explicitly indicated. Here and in the following figure, the convolution of β was performed using an FWHM of 1 eV. Also reported are some sets of experimental data: those of Parr et al. [21], with both the $v' = 0$ and $v' = 1$ sets; those recorded by Keller and co-workers [35] and those of Holland and co-workers [4]. In the upper section of the figure we report the unconvoluted asymmetry parameter (IC-RPA, Length Gauge).

located around 15 eV and found to have a narrower shape in the non convoluted results of Levine and Soven [71]. For energies greater than 20 eV both SC and IC results are close to experiment, thus confirming the quality of our calculations.

In Fig. 6 the presence of a second less pronounced minimum where the A_1 resonance appears at about 13 eV can be noticed. This has a correspondence in the data of Keller et al. [35], who found a narrow minimum of β at the same energy for the vibrational ground state of the $(1\pi_u)^{-1}$ parent ion. Both the vibrationally resolved data of Parr et al. [21]

and the unresolved data of Holland et al. do not display such a feature.

Further information can be gained by looking at Fig. 7, which permits an analysis of the partial cross section spectrum for the $(1\pi_u)^{-1}$ channel in terms of individual *pwc* contributions. In the SC approximation the (2,1) wave dominates unchallenged below 18 eV and then dies out, whereas the contribution of the (2,2) wave remains relatively constant, and thus it becomes the most important beyond 18 eV.

In the IC spectrum the (2,1) wave receives almost all the intensity from the A_1 resonance at 13 eV and the correlation of this maximum with the small dip of β around 13 eV obtained in the IC scheme is evident. The large effect of the A_1 resonance on the cross section and its smaller effect on β may be easily explained by considering that the (2,1) wave gives the largest contribution to the intensity around 13 eV, with or without the interactions with the resonance.

The (2,1) contribution has a second maximum in the 15–16 eV range arising from the interaction with $(3\sigma_g \rightarrow \sigma_u)$ Rydberg states. For the Π_u total symme-

try the interaction of the B_1 state with the (0,0), (2,0) and (2,2) waves results in a net maximum in the intensity at about 15 eV. These data are correlated with the main deep of β at about 15 eV.

In summary, it appears that, in spite of the apparent scattering and the absence of error bars in the experimental data, the location of the main minimum and roughly its main features are essentially reproduced by theory. Note also that an analysis of the vibrationally unresolved data [4] shows that if the branching ratios for each vibrationally resolved β are combined to obtain the unresolved asymmetry parameter, a shallower, broader minimum than seen for $v' = 0$, and which is slightly displaced to higher energies, e.g. even closer to that resulting in our calculations, would be obtained.

4.3. Photoionization from the $3\sigma_g$, $2\sigma_u$ and $2\sigma_g$ orbitals

The partial cross section for the process leading to the $(3\sigma_g)^{-1} A^2\Sigma_g^+$ ionic state is shown in Fig. 8, together with the corresponding asymmetry parameter.

Agreement with the (e,e + ion) measurements of Ref. [2] is in general quite good, with the interaction between the ionization channels slightly worsening the picture at high energy.

As also observed by some authors [48,49], the maximum at the threshold is due to the shape resonance corresponding to a $3\sigma_g \rightarrow \sigma_u^*$ transition. The assignment is based on the partial wave analysis for the A channel – analogous to that of Fig. 7 referred to the X channel – showing undoubtedly that the maximum is due to the (1,0) and (3,0) contributions. The whole spectrum corresponds to that seen in the $(3\sigma_g)^{-1}$ channel of N_2 , with a red energy shift in the shape resonance position with respect to molecular nitrogen.

The asymmetry parameter displays a neat minimum around 20 eV, a feature related to the presence of the shape resonance discussed just above, and which confirms the results of Ref. [48]. The data of Keller et al. [35], which lie in between our SC and IC predictions, do not extend to this region, and thus do not confirm the presence of this dip. The recent data of Holland and co-workers [4] on the other hand

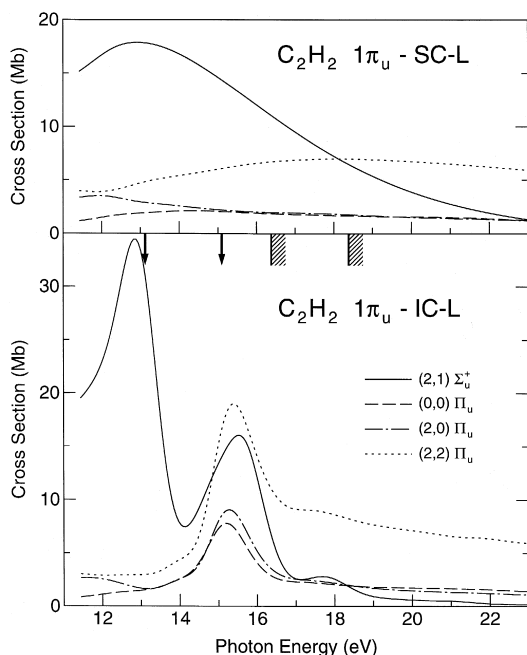


Fig. 7. Individual length (L) gauge (*l,m*) contributions to the partial $(1\pi_u)^{-1}$ photoionization cross section of C_2H_2 (Mb) in the 11 eV to 23 eV photon energy range.

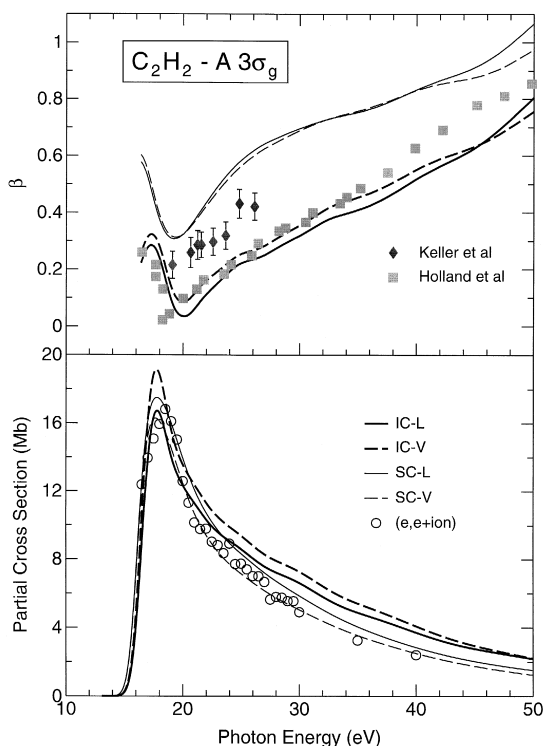


Fig. 8. The partial $(3\sigma_g)^{-1}$ photoionization cross section of C_2H_2 (Mb) in the 10 eV to 50 eV photon energy range (below). The asymmetry parameter β is shown in the upper section of the figure. Also reported are the experimental (e,e+ion) data extracted from Ref. [2] and the PES photoelectron angular distribution results of the measurements of Holland and co-workers [4].

show a well defined minimum at 19 eV, and they sit practically on top of our IC results for most of the energy interval shown in the Figure. This is thus another example of the crucial role of interchannel interaction in our study. As in Ref. [48] and [4], our asymmetry parameter increases uniformly in the 20 eV to 50 eV energy interval, and does not confirm the existence of the broad dip predicted by Thiel [72] and Farren et al. [49] and due to another shape resonance of Σ_u^+ ($3\sigma_g \rightarrow 4\sigma_u$) symmetry in the 35–40 eV interval. Lynch et al. [48] have shown that the eigenphase sum for the σ_u $A^2\Sigma_g^+$ ion, after a peak around 20 eV, increases again after 35 eV, a sign of the presence of this second shape resonance. They predict that a corresponding dip in the asymmetry parameter might go undetected in vibrationally unresolved spectra. It should be observed however that

the intensity due to the $3\sigma_g \rightarrow 4\sigma_u$ transition would be almost negligible [48].

Fig. 9 displays our results for the partial photoionization of acetylene for the process leading to the ion excited $(2\sigma_u)^{-1} B^2\Sigma_u^+$ state. Both SC and IC show a peak around 20 eV, followed by a decrease, far more pronounced in the IC approximation. The cross section is one order of magnitude smaller than exhibited by the $(1\pi_u)^{-1}$ ionization channel, and the scale of the figure magnifies the differences between SC, IC and the rather scattered experimental data extracted from Ref. [2]. Nonetheless it must be noted that SC, which displays the usual degree of worsening of the gauge agreement, seems to get closer to experiment than IC, where on the other hand results in the Length and Velocity choice of gauge are almost indistinguishable.

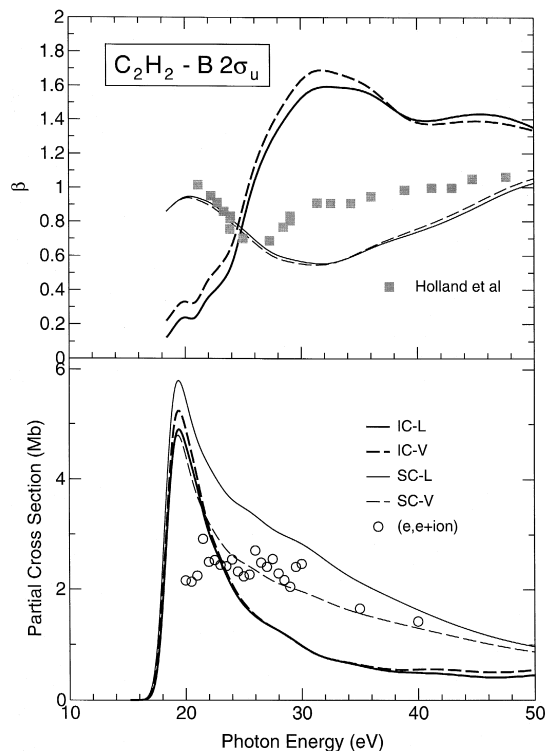


Fig. 9. The partial $(2\sigma_u)^{-1}$ photoionization cross section of C_2H_2 (Mb) in the 10 eV to 50 eV photon energy range (below). The asymmetry parameter β is shown in the upper section of the figure. Also reported are the experimental (e,e+ion) data extracted from Ref. [2] and the PES photoelectron angular distribution results of the measurements of Holland and co-workers [4].

As pointed out by Lynch et al. [48], the steep increase around 20 eV, which has apparently no counterpart in experiment, is related to the presence of intense discrete valence-like $2\sigma_u \rightarrow n\sigma_g$ transitions about 1 eV below the ionization limit, leading to the peak right after threshold. A similar occurrence can be observed in the $2\sigma_u \rightarrow n\pi_g$ manifold. The authors interestingly point out that no such behavior can be observed in the $(2\sigma_u)^{-1}$ partial cross section of N_2 , where the strong transitions of $2\sigma_u \rightarrow n\pi_g$ character appear more than 5 eV below the corresponding ionization threshold. The fine structure of the photoelectron spectrum of acetylene at the threshold for ionic states of interest here is given with plenty of details in the study of Avaldi and co-workers [3], who find for this channel a rapid decrease of the cross section soon after the threshold. No immediate explanation can be found for this.

The disagreements between SC and IC are strongly magnified in the asymmetry parameter. Where SC shows a (broad) minimum (around 30 eV), IC displays a maximum, and vice-versa at lower photon energies. To further complicate the analysis, experiment is better mimicked by the SC rather than the IC results. Here again Holland et al. [4] observed a minimum at about 25 eV, in agreement with the predictions of Lynch et al. [48], who explain it with the influence of the strong $2\sigma_u \rightarrow 4\sigma_g$ transition falling right below threshold. Our minimum in the SC approximation is shifted to higher energies by about 5 eV, and it seems to be much too broad, compared to experiment.

Fig. 10 shows our results for the partial photoionization of the inner-valence $2\sigma_g$ molecular orbital, leading to the $(2\sigma_g)^{-1} C^2\Sigma_g^+$ ion state.

It has been argued by several authors that the appearance of several satellites in the photoelectron spectrum [4] indicates in this case that the single particle model no longer holds.

Assignments of spectral lines based upon the one electron model is highly questionable in such a case [48]. No comparison to experiment seems to be available for the cross section, and we can only observe that our SC and IC are in qualitative agreement, with a minor peak at 25–27 eV and a steady increase after 35 eV, presumably leading to a second broader and more intense peak beyond 50 eV. The spectrum resembles that computed by Lynch et al.

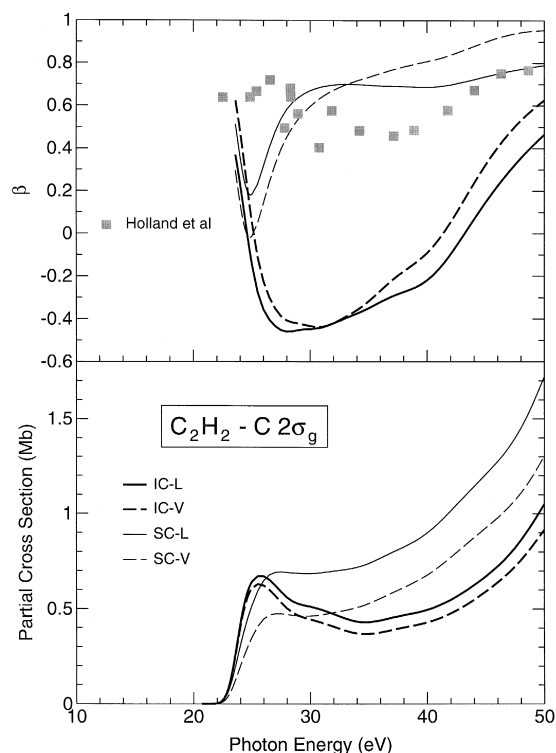


Fig. 10. The partial $(2\sigma_g)^{-1}$ photoionization cross section of C_2H_2 (Mb) in the 10 eV to 50 eV photon energy range (below). The asymmetry parameter β is shown in the upper section of the figure. Also reported in the upper section are the PES photoelectron angular distribution results of the measurements of Holland and co-workers [4].

[48], who observe the two maxima around 26 eV and 45 eV respectively. The authors argue that these enhancements in the cross section are due to a shape resonant behavior of the σ_u continuum ($2\sigma_g \rightarrow 3\sigma_u$ and $2\sigma_g \rightarrow 4\sigma_u$ respectively, see also Ref. [49]), in analogy to what discussed above for the $(3\sigma_g)^{-1}$ partial channel, but with the difference that in this case the effect of the broad resonance around 45–50 eV is made explicit in the spectrum. Note also that, this time as expected, the L-V agreement improves upon introduction of the interchannel interaction.

The behavior of the partial cross section reflects also on that of the asymmetry parameter β . Both SC and IC display a dip, a very sharp one and very close to the threshold for SC and a much more pronounced and broad for IC. The two approaches give rise thus to very different asymmetry parameter profiles. On

the other hand the major qualitative features are reproduced in both cases, and lead to some degree of agreement both with the theoretical predictions of Lynch et al. [48] – whose β essentially is reproduced in essence also quantitatively by our SC results – and also with experiment [4]. The latter has a value of about 0.7 at threshold, goes to a minimum of about 0.4 around 36 eV, and then rises smoothly. The minimum is broad but much less pronounced than our IC calculations predict. The data are scattered, their uncertainty is not given in the Figure. The authors of Ref. [4] seem to be unable to observe the dip around 26 eV predicted by the analysis of Lynch and co-workers, or, for that matter, the other dip around 46 eV, and they found only a shallow broad dip right in the middle of this region. Our Interacting Channel results seem to confirm these findings, again at least qualitatively.

5. Conclusions

We performed a detailed study of the features of the photoelectron spectrum of molecular acetylene. We employed the K-matrix L^2 approach which has been experimented with success in our group in the last decade, and exploited the advantages given by the use of Gaussian type orbitals for the description of the continuum partial waves. An analysis of the effect of interchannel coupling on the total and partial cross section and on the asymmetry parameter, including the three outer-valence and the inner-valence $2\sigma_g$ ionization channels, was carried out.

The major features of the photoelectron spectrum of C_2H_2 were recovered. The technique allowed to study in some detail the interplay of multichannel effects on the shape resonances and the subtleties of the intensity borrowing phenomena, especially evident in this system. For high photon energies, when the innermost shells are involved, the effect of electron correlation would require quite certainly the introduction of several double excitations of the outer valence shell in the basis. It is probable that their absence in our analysis might be the principal cause of the disagreement between the computed cross sections and asymmetry parameters and the corresponding experimental data for the $(2\sigma_u)^{-1}$ and $(2\sigma_g)^{-1}$ channels, as well as for the excess of oscil-

lator strength that we find in the lower portion of the energy spectrum.

The study of the photoelectron spectrum of acetylene can be considered another test for the theoretical approaches to the description of molecular continuum. In our analysis we're moving step by step toward more complex and more challenging systems. We began with the "simplest" molecule (H_2 [60]), then tackled Li_2 and LiH [61], whose experimental interest might be limited, but where most of the mechanisms of our technique were put under test. The study of molecular nitrogen [62] meant heavy computational effort and scrupulous analysis of multichannel effects, and bore an immediate interest both in connection with the large amount of experimental data and with the abundance of theoretical studies. Acetylene, although isoelectronic to nitrogen, bears the need of even more extended computational effort, and of a more accurate account of the behavior of the continuum orbitals, in particular in the regions where the nuclei, in this elongated molecular systems, exert a strong influence on the continuum wave functions. Although still some of the features of the photoelectron spectrum escape a neat description, we can safely state that the major effects are correctly described by our approach, and that the interaction between the excitation channels proves to be a determining factor to explain some of the finer and most interesting details.

Acknowledgements

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